

AI & ML for Data Scientists

Class 9: Regression Tree

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Introduction

Loss function

Finding Best Split

Pros and Cons

Applications

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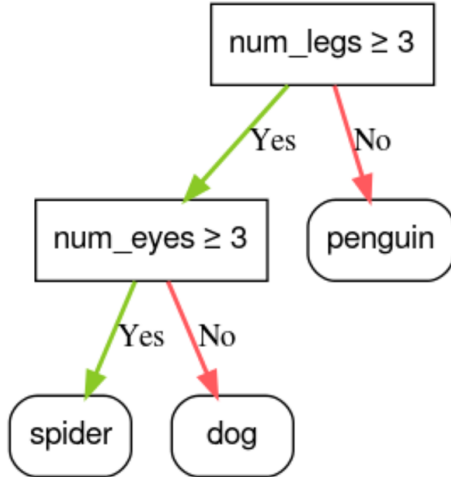
Pros and Cons

Applications

What is a Decision Tree?

- Decision Tree is a supervised learning model used for:
 1. Classification
 2. Regression
- It splits data into regions based on feature values, forming a tree-like structure:
 - Simple explanation: dividing the data into groups

Binary Decision Tree



A binary decision tree

Math of Decision Tree Algorithm

- $\hat{f}(X) = \sum_{m=1}^5 c_m I \{(X_1, X_2) \in R_m\}.$
- Equation from Element of Statistical Learning Page 305
- Just divide the data into 5 groups based on values of features X_1 and X_2 . The c_m is the predicted value of group m

Why Decision Tree Important

- It is basic element for the Random Forest and Gradient Boosting (xgboost)
- If you understand the Decision Tree well, you got 99% of RF and Xgboost

Introduction

Loss function

Finding Best Split

Pros and Cons

Applications

Key Ideas to minimize the loss function

- A decision tree splits the feature space into regions.
- At each split, choose the feature and threshold that Minimize loss

Decision Tree for Regression

- Use MSE, or MAE, or Huber, or ... (No difference with the linear regression)
- The MSE loss split minimizes:

$$\text{Cost} = \text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \bar{y})^2$$

Where $\bar{y}$ is the mean of target values in the group (node). ¹

¹Cost is the average loss across sample

Decision Tree for Classification

Common impurity metrics:

- Cross Entropy Loss (a.k.a log loss):

$$Cost = - \sum_{k=1}^K p_k \log(p_k)$$

- Gini Loss (Impurity, only for the decision tree algorithm):

$$Cost = 1 - \sum_{k=1}^K p_k^2$$

Where p_k is the proportion of class k at the group (node).

Loss Example for a Simple Node (Group)

Suppose a node (group) contains 10 samples:

6 (Positive), 4 (Negative)

Cross Entropy Loss:

$$\text{Cost} = -(0.6 \log 0.6 + 0.4 \log 0.4) \approx 0.97$$

Gini Loss (impurity):

$$\text{Cost} = 1 - (0.6^2 + 0.4^2) = 0.48$$

Cost of New Split

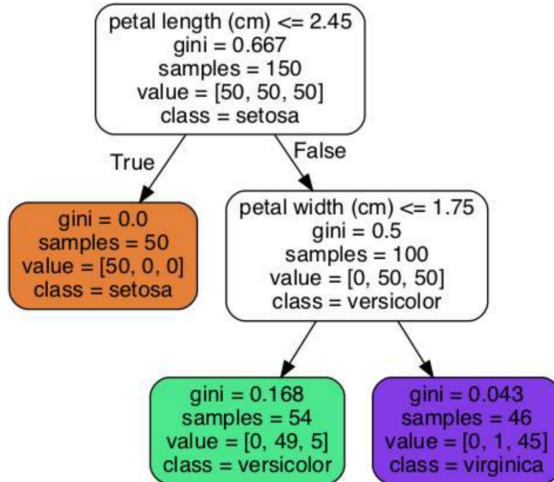
A new split divides parent group (node) P into children left group (node) L and right group (node) R .

$$Cost_{new} = \left(\frac{n_L}{n_P} Cost(L) + \frac{n_R}{n_P} Cost(R) \right)$$

Where:

- $Loss(\cdot)$ is loss (Entropy or MSE or Gini)
- n_L, n_R are sample counts for left group and right group. $n_P = n_L + n_R$, which is sample size of parent group.

Regression Tree: Full Picture



Introduction

Loss function

Finding Best Split

Pros and Cons

Applications

How to find the best split?

- Find the optimal split directly is hard (NP impossible)
- We turn to the greedy grid search to find the best split

Greedy Algorithm for Split Finding (Pseudocode)

- n is number of features, m is different values of feature n

Algorithm 1 Find best split

- 1: **Input:** dataset D
 - 2: **for** each feature n **do**
 - 3: **for** each candidate threshold θ_m for feature n **do**
 - 4: split D into $D_L = \{x_n \leq \theta_m\}$, $D_R = \{x_n > \theta_m\}$
 - 5: compute Loss
 - 6: **end for**
 - 7: **end for**
 - 8: partition D by best split with smallest loss
-

Greedy Algorithm (Simple)

Actually:

- Loop over feature (n) and loop over feature values (m) to get best (smallest) loss value²

²what is the time complexity here?

Example

Please follow to the blackboard

Avoid Over-fitting

To avoid over-fitting, tree grows until:

- Minimum samples per node
- Maximum depth reached

Introduction

Loss function

Finding Best Split

Pros and Cons

Applications

Advantages

- Model automatically has feature non-linearity and feature interaction (Why?)
- Handles missing values well
- Robust for feature scale differences.
- Intuitive and interpretable

Disadvantages

- Prone to overfitting
- Unstable (small changes $\rightarrow$ different trees)

From One Tree to Many Trees

- One decision tree may not perform well and has many disadvantage
- Solution: how about use many decision trees?
- Many decision trees → random forest & xgboost (we will study them in next class)

Introduction

Loss function

Finding Best Split

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Applications

Real-World Applications

- Credit approval
- Medical diagnosis
- Fraud detection
- Customer segmentation

End of Slides

Questions?

Reference

1. Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow (3rd edition)
2. Andrew Ng's Coursera
3. Kaggle
4. Wikipedia
5. ChatGPT
6. DeepSeek